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20. ABSTRACT (Continue on reverse side if necessary and identify by block number)

Describes methodology for evaluating system-peculiar test, measurement, and diagnostic equipment (TMDE), including physical and operational characteristics. Procedures are provided for initial inspection, physical characteristics, safety, performance, extreme environments, r-f interference, reliability, maintenance, and human factors. Supplementary instructions are provided for identifying the test item, documenting test criteria, developing performance tests, environmental tests, test plan organization, and maintenance evaluation of the TMDE.

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REFERENCE FILE

U. S. ARMY TEST AND EVALUATION COMMAND
DEVELOPMENT TEST II (EP) - SYSTEM TEST OPERATIONS PROCEDURE

AMSTE-RP-702-105
Test Operations Procedure 6-2-335
AD No. 781946

7 May 1974

TEST, MEASUREMENT, AND DIAGNOSTIC EQUIPMENT (SYSTEM PECULIAR)

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SECTION I
GENERAL

1. Purpose and Scope. This TOP provides guidance for planning tests of system-peculiar test, measurement, and diagnostic equipment (TMDE) to insure its conformance with Required Operational Capabilities/Development Plans (ROC's/DP's) and other guidance documents. Subtests to satisfy the requirements for the particular TMDE and test type (i.e., Development Tests (DT) II and III) can be selected or supplemented from those listed in section II.

2. Background. US Army field type systems are designed and issued with a wide variety of TMDE. This equipment is used to perform status evaluations, troubleshooting and repair actions, and requalification maintenance functions on the supported end items. Of the several basic categories of this equipment (i.e., common, special, and system-peculiar), the system-peculiar TMDE is of sole concern in this TOP.

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System-peculiar TMDE is equipment designed to support only one end item of Army materiel. There are two types: built-in test equipment (BITE) and separate test equipment. This equipment may be designed for use at one or more of the maintenance levels - operator, organizational, direct support, general support and depot - depending on the particular situation. For example, BITE is commonly used by the end item operator, and its maintenance may be performed by a combination of efforts by the operator and one or more of the other maintenance levels or by the operator and depot levels only.

BITE is normally evaluated concurrently with the DT II (EP) and DT III of the end item to which it is mounted. Separate TMDE is preferably evaluated concurrently with the test item, but if it is developed out of phase with the test item, evaluation may occur at a later time.

In test planning, the system-peculiar TMDE to be evaluated is considered to be one of the many elements of the maintenance support planned for a particular end item. Test planners must therefore consider not only the interface between the TMDE and the end item but also the TMDE interface with the other elements of the planned maintenance support such as the manuals, common test equipment and tools, repair parts, calibration facilities, etc.

Criteria for the test plan include not only the requirements stated in the ROC/DP and test directive but also applicable statements in the logistical support section of the DP. The test methodology and analytical plan must be provided in sufficient organization and detail to provide a systematic and comprehensive means for evaluating the equipment relative to each test criterion.

Proper testing of TMDE cannot be achieved (as has sometimes occurred in the past) if the TMDE is put to use only when there is a problem during the testing of the end item. To test the TMDE, certain faults must be planted in the end item to determine whether proper diagnosis can be achieved. Additionally, if the TMDE is to be taken into the field with its end item, it must be subjected to the same environmental conditions as the end item.

A study on testing TMDE is contained in reference 9 (app. A).

3. Equipment and Facilities. Equipment and facilities are covered in the content and references of section II.

SECTION II
TEST PROCEDURES

4. Supporting Tests. Applicable TOP's/MTP's, military standards and other documents, and the tests defined in section III (in preferred order of completion with respect to high-risk, short duration) to be considered in formulating a DT test plan are listed below. The section III test

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methods are written to provide broad guidance for planning an ET type evaluation of a specific TMDE design. It may be necessary to incorporate additional tests, modify some of the methods outlined, or consult other TOP's/MTP's. Whether or not separate test plans are used, the test of the TMDE will, when possible, be planned to be concurrent with the test of the end item, and, where practicable, subtests (e.g., salt fog test) will be conducted together. Test planning is discussed in paragraph 5.

	TEST SUBJECT TITLE	PUBLICATION NO.
a.	Initial Inspection (refer to para 6)	
b.	Physical Characteristics	1-2-504
c.	Safety Evaluation	3-2-503
d.	Performance Tests (refer to para 7)	6-2-090, 7-2-061 (when issued)
	(1) Self-Check Test (refer to para 7b)	
	(2) Purchase Description Tests (refer to para 7a)	
	(3) Operational Equipment Tests (refer to para 7c)	
	(4) Potential Fault Detection and Isolation Tests (refer to para 7d)	
	(5) Standard Design Characteristics (refer to para 7e)	
	(6) TMDE Interface Tests (refer to para 7f)	
e.	High Temperature Tests (refer to paras 8 and 9)	AR 70-38
f.	Low Temperature Tests (refer to paras 8 and 9)	AR 70-38
g.	Road and Cross-Country Travel (refer to para 8)	2-2-808, 1-2-601
h.	Laboratory Shock and Vibration (refer to para 8)	1-2-601; MIL-STD-810C, Methods 514.1 and 516.1
i.	Railroad Transportation (refer to para 8)	1-2-500
j.	Radio Frequency Interference (refer to para 8)	6-2-542

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	TEST SUBJECT TITLE	PUBLICATION NO.
k.	Rain (refer to para 8)	MIL-STD-810C, Method 506; 2-2-815, 6-2-533
l.	Transit Drop (refer to para 8)	MIL-STD-810C, Method 516.1
m.	Immersion (refer to para 8)	2-2-612; MIL-STD-810C, Method 512
n.	LOTS (refer to para 8)	2-2-510 (when issued)
o.	Air Transportability (refer to para 8)	1-2-500
p.	Solar Radiation (refer to para 8)	4-2-826
q.	Reliability	6-2-503, 5-1-014
r.	Maintenance Evaluation (refer to para 10)	1-2-501 (when issued)
s.	Human Factors Evaluation	5-2-545, 6-2-502
t.	Sand and Dust (refer to para 8)	MIL-STD-810C, Method 505
u.	Humidity (refer to para 8)	4-2-820, 6-2-534
v.	Salt Fog (refer to para 8)	MIL-STD-810C, Method 509
w.	Fungus (refer to para 8)	4-2-818; MIL-STD-810C, Method 508

SECTION III
SUPPLEMENTARY INSTRUCTIONS

5. Test Planning. The test planner must be thoroughly familiar with the stated Army requirements for the end item as stated in the ROC/DP (including the logistical support section of the DP), engineering design handbooks and other technical documents. He must also be familiar with the characteristics of the planned interface between the TMDE and the end item and the other elements of the maintenance support planned for the end item through general support (GS) level maintenance. The maintenance level(s) where the TMDE is to be employed and the particular conditions of extreme operating and storage environments planned for the TMDE must be known. The evaluation of system-peculiar TMDE may be planned as a part of an evaluation of the overall system (i.e., the end item and all its supporting equipment) or separated from the end item and treated in a separate test plan.

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The evaluation of system peculiar TMDE as a part of an overall system evaluation should include special precautions to insure that the planned TMDE evaluation is not obscured by the end item evaluation. For example, the TMDE evaluation may be organized as part B of section II of the test plan (part A would comprise the end item subtests), the TMDE evaluation may be included as subelements of one or more of the required end item subtests, or some other technique may be used that is within the guidance of TECR 70-24. The test plan should include background information and methodology relative to the TMDE in the following areas:

a. The description of materiel should identify and describe the requirements and major characteristics of the planned system-peculiar TMDE to be employed through GS level. The technical aspects of the maintenance support plan for the supported end item and the system-peculiar TMDE should also be summarized through GS level.

b. When documenting test criteria, the test planner should consider not only criteria sources such as ROC's/DP's, AR's, and the test directive, but also statements in the logistical support sections of DP's that are considered to be criteria applicable to the planned evaluation. Emphasis should be placed on specific performance requirements for the TMDE, such as the probabilities that (1) an operational item will actually be shown to be operational, (2) a defective item will actually be shown to be defective, (3) the faulty item requiring repair action will be correctly identified, and (4) correct repair instructions will be provided.

c. Systematic test methodology should be developed to determine the necessity and adequacy of the TMDE performance for all the required maintenance tasks through GS level.

d. Test methodology should be developed to determine the adequacy of the TMDE to meet the full range of environmental requirements applicable to its operation and storage. In many instances the TMDE requirements will differ from the supported system.

e. Test methodology should be developed to determine the adequacy of the TMDE to meet other test requirements for initial inspection, physical characteristics, safety, reliability, and maintainability.

6. Initial Inspection.

a. Literature pertinent to the TMDE, including its maintenance test package and other end item support equipment essential to a TMDE evaluation, is reviewed for familiarization with performance requirements, operational characteristics, the functions of components, and the results of previous tests. The literature includes the approved test plan, drawings, draft and final equipment publications, the safety statement, and reports of previous tests.

b. The end item and the TMDE, including its maintenance test package, are inspected for damage which, if present, should be investigated as to cause, reported by EPR, and corrected before the start of the test.

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c. The TMDE, including its maintenance test package, is inventoried for completeness. If the complete TMDE has not been received, a teletype EPR will be forwarded to the test sponsor and other agencies in accordance with AMCR 700-38. In this event, tests will not commence until either the missing materiel arrives, or a waiver is received from Headquarters, AMC.

7. Performance. Performance tests are conducted to determine whether the system-peculiar TMDE is necessary and adequate to support the end item in accordance with the ROC/DP and logistical support section of the DP. The tests may include a theoretical engineering study of one or more characteristics of the TMDE to provide supplemental data when actual hardware tests must be unduly limited for some reason.

a. Purchase Description Tests. The TMDE is subjected to specification performance tests to insure that it is operating within prescribed limits and that it is a valid sample for evaluation. Measured parameters are compared with specified values. Go, No-Go indicators are verified using specified Go, No-Go signals and marginal signals.

b. Self-Check Test. The TMDE is subjected to its specified self-checks and other technical manual preoperational checks to determine whether it will meet its own criteria. Self-check validity is verified by comparing self-check results with results obtained from using independent instrumentation when the unit is operational and when potential faults are introduced (see d below for a technique for introducing faults).

c. Operational Equipment Tests. The TMDE is tested to determine whether it will indicate that an operational system is operational (i.e., not indicate faults where none exist). In this test the system parameters are checked with independent instrumentation to insure that it is within purchase description requirements. The "calibrated" system is then checked with the TMDE and any discrepancies are noted.

d. Potential Fault Detection and Isolation Test. The TMDE is tested to determine whether it can detect and isolate faults as appropriate and as required by the ROC/DP, logistical support section of the DP, etc. Potential faults are identified for insertion into the supported system to provide the TMDE test problem. A 100% sampling plan is used when the number and/or characteristics of potential system faults are not too great for test tiem and cost considerations. When a 100% TMDE checkout is considered too time consuming or costly, a sequential sampling plan is designed. A proposed sampling plan and a hypothetical example are shown in appendix B. The following conditions must be satisfied when using a sequential sampling plan:

(1) The test faults must be a random sample of the total population of potential faults.

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(2) The order of test fault insertion must be random.

The following precautions should be considered before inserting faults:

(1) Potential system faults that would damage the TMDE should not be inserted.

(2) Potential system faults that would damage the system should not be inserted.

Certain potential faults of particular interest which are not inserted during the sequential sampling test should be scheduled for evaluation during a special supplemental test. The identification of the total population of potential faults is accomplished by itemizing each individual piece or part (i.e., resistor, capacitor, transistor, etc.) and the various related failure modes (i.e., shorts, opens, grounds, biased values, etc.). If this task is considered too time consuming and costly, an alternate approach is to designate potential faults in each of the subassemblies or at some other convenient level until the sample is adequate for the sequential sampling plan being used.

e. Standard Design Characteristics. This test determines whether the TMDE is designed and configured in accordance with standard Army requirements. The TMDE characteristics are checked against the criteria in the checklist (fig. 2) of appendix C (from AMCP 706-134) and any discrepancies are noted. Other checklists of AMCP 706-134 should be added if considered appropriate.

f. TMDE Interface Tests. This test is conducted to determine any engineering type discrepancies between the TMDE and the other elements of maintenance support. The TMDE should be reviewed in the following areas:

(1) Could common or other equipment in Army inventory be adapted to satisfy the user's requirement in lieu of the system-peculiar TMDE under evaluation?

(2) Is the system-peculiar TMDE technically compatible with the common TMDE planned for the system?

(3) Are the technical manuals technically adequate for using the system-peculiar TMDE?

(4) Are the planned calibration facilities technically adequate to calibrate the system-peculiar TMDE in all areas?

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8. Extreme Environments. Extreme-environment tests are conducted to determine whether the performance of the system-peculiar TMDE is degraded by exposure to the various extreme environmental conditions expected in field use. The following two factors are of particular concern in planning these tests:

a. Each environmental exposure must be appropriate to the particular TMDE.

b. The scope of checks to detect TMDE physical and operational degradation must be adequate.

The environmental conditions planned for the TMDE often differ significantly from those expected for the end item, often being less severe but never more severe. The environments for organizational, DS, and GS levels of TMDE also often differ significantly from each other. The test planner must therefore review the ROC's/DP's, the logistical support sections of the DP's, AR 70-38, and other documents to clearly identify the field conditions appropriate to the TMDE. Once the field conditions are identified, the development of the test exposure conditions (i.e., miles, temperatures, etc.) should be accomplished using standard planning practices identified in the TOP's, military standards, etc., of paragraph 4. The scope of physical and operational TMDE tests planned to be conducted before, during, and after each exposure is often abbreviated from the performance tests of paragraph 7 because of time and cost factors. In this respect there are at least two factors that should be taken into consideration when establishing the scope of the TMDE checks:

a. The scope of TMDE checks should at least equal the scope of checks for systems of equal complexity.

b. The scope of TMDE checks should reflect (1) the importance of TMDE performance in checking major performance parameters of the supported system and (2) the characteristics of the particular environment.

9. High and Low Temperatures. Both the high and low temperature tests have a storage phase and an operating phase. Unless specifically designated otherwise, the high temperature storage and operating tests and the low temperature storage tests of TMDE will be conducted at the same temperatures and for the same durations as those of the end item. The low temperature operating test for TMDE may be more moderate than that of the end item and will be in accordance with the ROC/DP or logistical support section of the DP.

10. Maintenance Evaluation. This test is conducted to determine whether the system-peculiar TMDE can meet specified maintenance requirements. It is particularly important to insure that this evaluation addresses the maintenance characteristics of the TMDE and the adequacy of its maintenance support elements as compared to the end item characteristics.

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APPENDIX A
REFERENCES

1. AR 70-38, "Research, Development, Test, and Evaluation of Materiel for Extreme Climatic Conditions."

2. AR 750-1, "Army Materiel Maintenance Concepts and Policies."

3. AR 750-43, "Test, Measurement and Diagnostic Equipment."

4. AMCR 700-38, "Test and Evaluation of Materiel - Correction of Defects Found During Materiel Life Cycle Testing."

5. AMCR 750-15, "Integrated Logistic Support."

6. AMCP 706-134, "Maintainability Guide for Design" (Engineering Design Handbook).

7. MIL-STD-810C, "Environmental Test Methods."

8. TECOM Supplement 1 to AR 750-1, "Army Materiel Maintenance Concepts & Policies."

9. Roberts, William M., "Special Study of Methodology for Evaluating System-Peculiar Test Equipment," Aberdeen Proving Ground, Md., Report APG-MT-4224, TECOM Project 9-CO-001-000-084, September 1973.

10. Wald, A., "Sequential Analysis," John Wiley and Sons, New York, N.Y.

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APPENDIX B
PROPOSED SAMPLING PLAN FOR DETERMINING ABILITY OF SYSTEM-PECULIAR TMDE TO DETECT POTENTIAL FAULTS

Component faults are intentionally programmed into a system and observations are made as to whether the TMDE can detect those faults. The sampling plan below is based on a sequential probability ratio test when the underlying parameter is binomial. (For further information see ch. 5, Wald, ref. 10, app. A.) This sampling concept may be adapted for use in other TMDE performance tests of paragraph 7 of this TOP.

A system that is to be diagnosed and, perhaps, maintained will have component faults or failures occurring during testing. The types of faults can be many and the numbers extremely large. The sampling plan is based on the assumption that the probability of incorrect indication (p) is constant from trial regardless of which fault occurs. Faults should be selected by some random process.

To construct a test of hypothesis two values of p must be selected. Let the fraction implying good TMDE equipment be denoted by p_o and let the fraction implying bad TMDE be denoted by p_1 ($p_1 > p_o$). If the true p is p_o or smaller, the risk of rejecting the hypothesis ($p = p_o$) is α or less (type I error); if the true p is p_1 or larger, the risk of accepting the hypothesis is β or less (type II error).

A sampling plan satisfying the conditions that the probability of rejecting proper detection of faults does not exceed α whenever $p \leq p_o$, and the probability of accepting proper detection of faults does not exceed β whenever $p \geq p_1$, is given by the sequential probability ratio test of strength (α, β) for testing the hypothesis $p = p_o$ against the hypothesis $p = p_1$. To carry out the test, the acceptance number a_k and the rejection number r_k are calculated (they depend only on p_o, p_1, α, β, k and can be calculated prior to actual testing). The process of fault injection is continued as long as $a_k < x_k < r_k$ where x_k denotes the number of failures (failure to properly detect a fault) in the k trials. The first time that x_k does not lie in the interval (a_k, r_k), the fault injection is terminated. If $x_k \geq r_k$, the hypothesis is rejected; if $x_k \leq a_k$, it is accepted. The acceptance number and the rejection number are given below.

$$a_k = \frac{\operatorname{Ln}\left(\frac{\beta}{1-\alpha}\right)}{\operatorname{Ln}\left(\frac{p_1}{p_o}\right) - \operatorname{Ln}\left(\frac{1-p_1}{1-p_o}\right)} + k\,\frac{\operatorname{Ln}\left(\frac{1-p_o}{1-p_1}\right)}{\operatorname{Ln}\left(\frac{p_1}{p_o}\right) - \operatorname{Ln}\left(\frac{1-p_1}{1-p_o}\right)}$$

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$$r_k = \frac{\text{Ln}\left(\frac{1-\beta}{\alpha}\right)}{\text{Ln}\left(\frac{p_1}{p_o}\right) - \text{Ln}\left(\frac{1-p_1}{1-p_o}\right)} + k\,\frac{\text{Ln}\left(\frac{1-p_o}{1-p_1}\right)}{\text{Ln}\left(\frac{p_1}{p_o}\right) - \text{Ln}\left(\frac{1-p_1}{1-p_o}\right)}$$

k = 1, 2,···, until decision is reached or until truncation.

These can be calculated before the test and prepared in tabular or graphical form for quick reference during the conduct of the test.

The sequential sampling plan does not provide any definite upper limit for the number of faults N to be programmed. Any large value of N is possible, but the probability is small that N will exceed twice or three times its expected value.

It is sometimes desirable to set a definite high upper limit N_o for N. This can be done by truncating the sequential process at $N = N_o$ where N_o is approximately three times the maximum expected value of N. Although the truncation process alters the risks, this effect is negligible for practical purposes when N_o is chosen as described above. The following is a reasonable rule for deciding acceptance or rejection at $N = N_o$ if no decision is reached for $N \leq N_o$ with the regular sequential procedure:

If $x_{N_o} < 1/2(a_{N_o} + r_{N_o})$, accept;

If $x_{N_o} \geq 1/2(a_{N_o} + r_{N_o})$, reject.

Additionally, if x_{N_o} failures occur prior to N_o, reject immediately.

The expected value of N depends on the fraction of programmed component failures that fail to be detected. The maximum expected value of N usually occurs at approximately

$$E(N) = \frac{\text{Ln}\left(\frac{\beta}{1-\alpha}\right) \cdot \text{Ln}\left(\frac{1-\beta}{\alpha}\right)}{\text{Ln}\left(\frac{p_1}{p_o}\right) \cdot \text{Ln}\left(\frac{1-p_1}{1-p_o}\right)}, \quad \text{thus } N_o = 3\ E(N)$$

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Example:

Suppose the hypothesis p = .10 against the hypothesis p = .15 is to be tested. Further suppose the α risk and β risk are taken as 0.10 and 0.20, respectively. The truncation number can be determined as follows:

$$N_o = 3 \cdot \frac{\text{Ln}\left(\frac{\beta}{1-\alpha}\right) \cdot \text{Ln}\left(\frac{1-\beta}{\alpha}\right)}{\text{Ln}\left(\frac{p_1}{p_o}\right) \cdot \text{Ln}\left(\frac{1-p_o}{1-p_1}\right)}$$

$$= 3 \cdot \frac{\text{Ln } (2/9) \text{ Ln } (8)}{\text{Ln } (1.5) \text{ Ln } (.85/.90)} = 3 \ (135) = 405$$

The acceptance and rejection numbers are calculated from the linear relations

$$a_k = -\ 3.251 + .124\ k$$

$$r_k = \quad 4.495 + .124\ k.$$

This plan is represented graphically in figure 1.

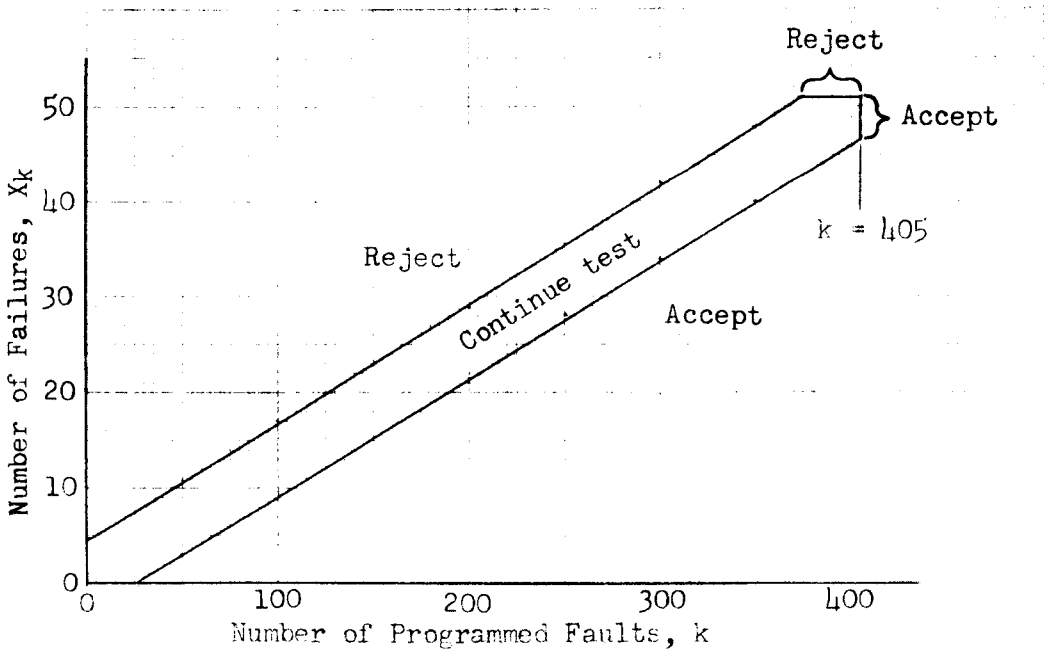


Figure 1. Sample Plot of Acceptance-Rejection Criteria.

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APPENDIX C
TEST EQUIPMENT CHECKLIST
(From AMCP 706-134)

1. Are the instructions for using test equipment in step-by-step format?
2. Is a signal provided which shows when the test equipment is warmed up?
3. If it is not feasible to present such a signal, is the warm-up time required clearly indicated near the warm-up switch?
4. Is a simple check provided to indicate when the test equipment is out of calibration or is otherwise not functioning?
5. Is appropriate indication of test equipment performance provided so the technician does not attempt to measure with a faulty standard or instrument out of calibration?
6. Do test equipment displays which require transformation of values have conversion tables attached to the equipment with the transform factor by each individual switch position or display scale?
7. Is adequate support provided for test equipment which must be taken into the work area so the technician does not have to hold the test equipment or take separate support devices to the work area for this purpose?
8. Are built-in test features provided wherever standard portable test equipment cannot be used?
9. Does portable test equipment packaging reflect the manner in which the equipment will be carried (i.e., size, shape, e.g., location of hand grips, clearance of technician's leg and of the floor, etc.)?
10. Does portable test equipment weigh under 14 lb. if it is to be carried by one man?
11. Do plugs, jacks and binding posts used for test of test equipment appear on outer casing of equipment so it is not necessary to remove the case? If internal repair requires removal of case, are duplicate jacks, plugs, etc. provided on chassis so jury-rig connections to the case are not necessary?
12. Are display lights, automatic power switches, or printed warnings provided to insure that test equipment is turned off when testing is completed?
13. Is storage for cable and test leads (within test instrument case or lid) designed so loose cable cannot interfere with closure of case?
14. Is purpose of test equipment and special cautions displayed in a conspicuous place on the outer surface of the test equipment?
15. Are units which are not self checking designed to be checked in the operating condition without the aid of special rigs and harnesses wherever possible?
16. Are selector switches provided in lieu of a number of plug-in connectors?
17. Is test equipment designed so as to be capable of connection to prime equipment within two minutes?

Figure 2. TMDE Checklist.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

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